

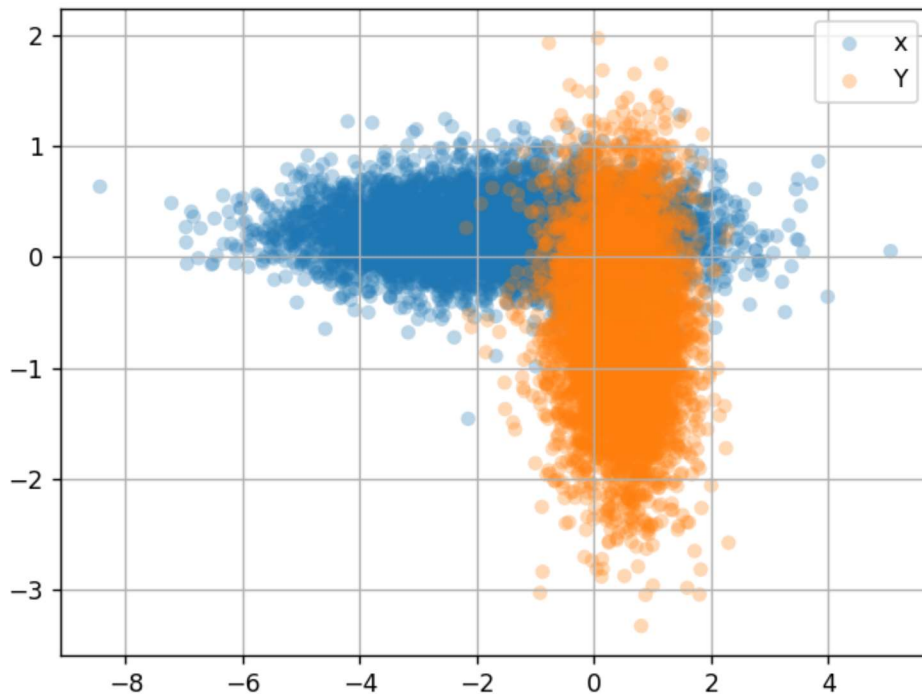
You are given, as the train data, `trn_x` and `trn_y` along with their class labels `trn_x_class` and `trn_y_class`. The task is to classify the

(a) classify instances in `tst_xy`, and use the corresponding label file `tst_xy_class` to calculate the accuracy;

(b) classify instances in `tst_xy_126` by assuming a uniform prior over the space of hypotheses, and use the corresponding label file

(c) classify instances in `tst_xy_126` by assuming a prior probability of 0.9 for Class x and 0.1 for Class y, and use the corresponding

Plot of the data:



Exercise A

The accuracy is found to be:
90.23%

An uniform prior of 0.5 has been chosen, as we do not intuitively know anything about the dataset, and therefore must assume that the data is equally split

Exercise B

The accuracy is found to be:
92.70%

With the new dataset, the uniform distribution is a little better

Exercise C

The accuracy has improved by 3.85% with a non-uniform distribution of 0.9 for x and 0.1 for y.

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